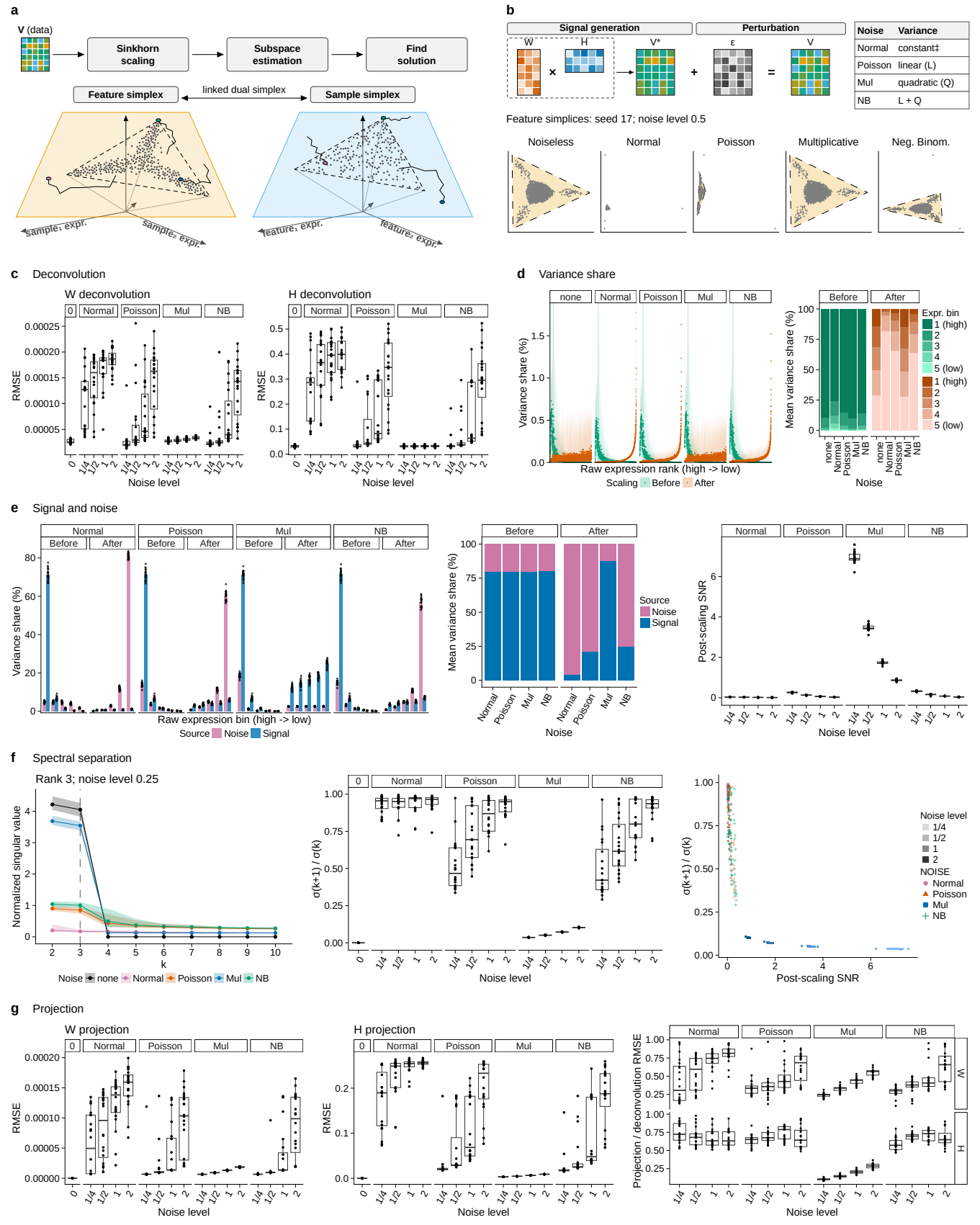

a
V (data)
Sinkhorn scaling
Subspace estimation
Find solution
Feature simplex
linked dual simplex
Sample simplex
sample₁ expr.
sample₂ expr.
feature₁ expr.
feature₂ expr.
b
Signal generation
Perturbation
W
H
V*
ε
V
Noise
Variance
Normal
constant‡
Poisson
linear (L)
Mul
quadratic (Q)
NB
L + Q
Feature simplices: seed 17; noise level 0.5
Noiseless
Normal
Poisson
Multiplicative
Neg. Binom.
c
Deconvolution
W deconvolution
H deconvolution
RMSE
Noise level
d
Variance share
Variance share (%)
Raw expression rank (high -> low)
Scaling
Before
After
Mean variance share (%)
Expr. bin
1 (high)
2
3
4
5 (low)
Noise
e
Signal and noise
Variance share (%)
Raw expression bin (high -> low)
Source
Noise
Signal
Mean variance share (%)
Post-scaling SNR
Noise level
f
Spectral separation
Rank 3; noise level 0.25
Normalized singular value
k
σ(k+1) / σ(k)
Post-scaling SNR
Noise level
NOISE
Normal
Poisson
Mul
NB
g
Projection
W projection
H projection
RMSE
Noise level
Projection / deconvolution RMSE